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SYNCLAB project proposal

Centralized Workshop & Laboratory Allocation Platform

Prepared 07 October 2026 from the existing program. This proposal-format supporting document does not establish any earlier proposal submission or initial presentation.

Problem and academic context

Shared university laboratories and workshops need a clear way to record student projects, review requests and prevent active reservations from occupying the same period. Facility, date, supervisor and project details should remain available during review and later changes. SYNCLAB applies graphical interface design, relational SQL data and C++ object-oriented methods to that academic scheduling problem.

Objectives

• Guide students through facility selection, date/slot choice and validated information entry.

• Store student profiles, workshop ownership, requests and selected periods in a normalized local SQLite database.

• Prevent overlapping active requests through transactional availability checks.

• Support key-based student editing and administrator status/remarks/filtering/deletion.

• Demonstrate encapsulation, inheritance and virtual-method polymorphism in integrated C++/Qt code.

Scope and intended users

Students create and revise requests. Workshop administrators review records, filter dates and maintain status/remarks. A system administrator manages local passcodes and appearance. The prototype covers ten workshop categories and four fixed daily periods. It uses a single local database; email actions prepare external drafts.

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Planned system components

The planned component description below is aligned with the supplied implementation. It presents the proposal structure without implying a newly approved or historical plan.

Component	Responsibility / technology
Interface	Qt Widgets / Designer pages: Welcome, Selection, Calendar, Form and Admin. Guided navigation and validation feedback.
Domain and rules	StudentRecord holds encapsulated values. AllocationRules validates fields and prepares allocation keys/status.
Persistence interface	IDatabaseManager declares virtual CRUD, filtering and availability operations.
SQL implementation	SqliteDatabaseManager uses prepared SQL and write transactions. Four tables: students, workshops, records and booking_slots.
Configuration / utilities	QSettings holds passcodes/theme/email configuration; storage and catalog helpers select paths/workshops.
Evidence and delivery	Source, runnable Windows bundle, ER/class/sequence diagrams, five SQL reports, test logs, report/manual and presentation.

Principal workflow

Welcome > workshop > calendar/free slots > form > save Pending and key. A saved key loads an edit. A workshop passcode opens administrator review. Rejected records release slots; reapproval checks conflicts. Joined SQL queries support submitted workload/utilization summaries.

Boundaries

There is no student account registration, network synchronization, built-in PDF attachment/report export or automatic email delivery. These features are future candidates. Passcodes and four-digit keys are lightweight prototype controls.

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Group roles and delivery plan

The information sheet requires five students and recommends the roles below. Names, index numbers, leader, actual contributions and presenter assignments await confirmation by the group. These are proposed responsibilities, not a declaration of work already performed.

Member identity	Proposed role	Responsibility
Name/index to confirm	UI/UX Designer	Screen flow, forms, styling and screenshots
Name/index to confirm	Database Designer	ER model, schema, SQL and data checks
Name/index to confirm	OOP Developer	Domain, rules, interface and class relationships
Name/index to confirm	Integrator	Connect GUI, rules, persistence and delivery
Name/index to confirm	Documentation & Testing Lead	Test evidence, report/manual and presentation

Milestones and verification

Initial stage: submit the 2-3-page proposal and deliver the mandatory five-minute overview. Prior completion is unverified. Final stage: provide executable/source, UI/UX documentation, schema/scripts and a 10-12-page report; present for ten minutes with live demonstration and all group members participating. Prepare for Week 14 pending course-date confirmation.

Success checks

Demonstrate a validated new request, key-based edit, administrator filter/status/remarks, conflict handling and deletion. Retain observed test results and meaningful SQL outputs. Confirm startup/display on the assessment computer and complete verified member attribution before hand-in.

Initial presentation material: G2. Final technical documents and executable: A-F. Real group details must replace the unassigned identities before submission.